



*Time  
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*ARIMA*

# *Time Series Analysis*

B. Banerjee



## ■ Text Book

- (1) Shumway, R. H., & Stoffer, D. S. (2017). Time series analysis and its applications: with R examples. Springer.
- (2) Brockwell, P. J., & Davis, R. A. (2016). Introduction to time series and forecasting. springer.

## ■ Reference book

- (1) Cryer, J. D., & Kellet, N. (1991). Time series analysis. Royal Victorian Institute for the Blind. Tertiary Resource Service.
- (2) Box, G. E., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). Time series analysis: forecasting and control. John Wiley & Sons.



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## *Introduction*



# Time series

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## Definition

Time series is a collection of random variables  $\{X_t \mid t \in T\}$  over a time index set  $T$ , which might be a finite, countably infinite or uncountable set.

- Realized values: What we observe are the realized values of the time series i.e. the data set is  $\{X_1 = x_1, X_2 = x_2, \dots, X_n = x_n\}$ , where the  $x_i$ s are some numeric or categorical values.



# Categories of Time series

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- Discrete time series: If  $T$  is a countable set then it is a discrete time series.
- Continuous time series: If  $T$  is an interval then it is a continuous time series.
- Note: Discrete or continuous are the adjectives of time but NOT of random variable  $X_t$



# Operators

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- Back-shift operator  $B$ , such that  $B^h X_t = X_{t-h}$
- Difference operator:  $I - B = \nabla$  which gives

$$\nabla X_t = X_t - X_{t-1} = (I - B)X_t$$

which implies

$$\nabla^h X_t = (I - B)^h X_t$$

- Seasonal difference

$$\nabla_s X_t = (1 - B^s)X_t$$



## Definition

The time series generated from uncorrelated variables with zero mean and fixed finite variance is called white noise [Notation  $W_t \sim WN(0, \sigma_w^2)$ ]

- Example1:  $X_t = W_t = \begin{cases} N(0, 1) & \text{if } t \text{ is even,} \\ \text{Exp}(1) - 1 & \text{if } t \text{ is odd.} \end{cases}$
- Example2 :  $X_t = W_t \text{ i.i.d. } N(0, \sigma^2)$



## Definition

A deterministic pattern ( $T_t$ ) which persists throughout in the time series. For example  $X_t = T_t + W_t$

- Example:  $X_t = T_t + W_t = 0.2t + W_t$
- Example:  $X_t = T_t + W_t = t^{(1/3)} + W_t$





## Definition

A deterministic pattern ( $S_t$ ) which returns in the time series after a fixed interval.  
For example  $X_t = T_t + S_t + W_t$

- Example:  $X_t = T_t + S_t + W_t = 0.3t + 3 \sin(4.5t\pi) + W_t$
- Example:  $X_t = S_t + W_t = 3 \sin(4.5t\pi) + W_t$



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## *Stationarity*



# Mean and Autocovariance

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## Definition

Suppose that  $\{X_t\}$  is a time series with  $E|X_t| < \infty$ .

Its mean function is

$$\mu_t = E[X_t]$$

## Definition

Suppose that  $\{X_t\}$  is a time series with  $E[X_t^2] < \infty$ , then its autocovariance function (ACVF) is

$$\gamma_X(s, t) = \text{Cov}(X_s, X_t) = E[(X_s - \mu_s)(X_t - \mu_t)]$$



# Weakly Stationary Time series

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## Definition

A time series  $\{X_t\}$  is said to be weakly stationary if

1.  $\mu_t$  is independent of  $t$ , and
2. For each  $h \in \mathbb{Z}$ , ACVF  $\gamma(t+h, t)$  is independent of  $t$

■ Example:  $X_t \sim WN(0, \sigma_w^2)$

■ NOT an Example:  $X_t = \sum_{i=0}^t W_i$ , where  $W \sim WN(0, \sigma_w^2)$



## Definition

$\{X_t\}$  is strongly stationary if for all  $k, h, t_1, \dots, t_k, x_1, \dots, x_k$  shifting the time axis does not affect the distribution i.e.

$$P(X_{t_1} \leq x_1, \dots, X_{t_k} \leq x_k) = P(X_{t_1+h} \leq x_1, \dots, X_{t_k+h} \leq x_k)$$

- Example:  $X_t \sim WN = N(0, \sigma^2)$
- NOT an Example: Random walk defined as

$$S_t = \sum_{i=0}^t X_i, \quad \text{where } X_i \text{ i.i.d. } N(0, \sigma^2)$$

- Strong Stationarity  $\implies$  Weak Stationarity



# Properties of a Strongly Stationary Time Series $X_t$

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- (1) The random variables  $X_t$  are identically distributed.
- (2)  $(X_t, X_{t+h}) \stackrel{d}{=} (X_1, X_{1+h})$  for all integers  $t$  and  $h$ .
- (3)  $X_t$  is weakly stationary if  $E(X_t^2) < \infty$  for all  $t$ .
- (4) Weak stationarity does not imply strongly stationarity.
- (5) An i.i.d. sequence is strongly stationary.



An important class of weakly stationary time series:

## Definition

$$X_t = \mu + \sum_{j=-\infty}^{j=+\infty} \psi_j W_{t-j}$$

where  $\mu \in \mathbb{R}$  and  $\sum_{j=-\infty}^{j=+\infty} |\psi_j| < \infty$  and  $W_j \sim WN(0, \sigma_w^2)$

- $E(X_t) = \mu$
- $\gamma_X(h) = \sigma_w^2 \sum_{j=-\infty}^{j=+\infty} \psi_j \psi_{j-h} < \infty$



## *WN as a linear process*

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- Time series  $X_t = W_t$  where  $W_t \sim WN(0, \sigma_w^2)$
- WN process is a linear process with  $\mu = 0$  and

$$\psi_j = \begin{cases} 1 & \text{if } j = 0 \\ 0 & \text{otherwise} \end{cases}$$

- WN had fixed and finite mean and variance with correlation zero
- WN need not be normally distributed
- WN need not be iid
- WN is always weakly stationary
- Normally distributed WN is strongly stationary
- iid sequence is always white noise and strongly stationary





## Theorem

Consider a (weakly) stationary time series  $\{X_t\}$  with mean zero and define

$$Y_t = \mu + \sum_{j=-\infty}^{j=+\infty} \psi_j X_{t-j}$$

where  $\mu \in \mathbb{R}$  and  $\sum_{j=-\infty}^{j=+\infty} |\psi_j| < \infty$  then

$$E(Y_t) = \mu$$

$$\gamma_Y(h) = \sum_{k=-\infty}^{k=+\infty} \sum_{j=-\infty}^{j=+\infty} \psi_k \psi_j \gamma_X(h-j+k) \text{ if exists.}$$



# Auto correlation function (ACF)

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Suppose that  $X_t$  is at least weakly stationary time series with

$$E(X_t) = \mu \text{ and } \gamma_X(h) = E[(X_t - \mu)(X_{t+h} - \mu)]$$

## Definition

Auto correlation function (ACF) of  $X_t$  and  $X_{t+h}$  is defined as

$$\rho(h) = \frac{\gamma_X(h)}{\gamma_X(0)}$$

- $\rho(h) = \rho(-h)$
- Let  $1 \leq i, j \leq n$  and define a matrix  $R = ((\rho(|i-j|)))_{i,j}$  then  $\mathbf{a}^T R \mathbf{a} \geq 0$  for all  $\mathbf{0} \neq \mathbf{a} \in \mathbb{R}^n$ .
- $R$  is positive semidefinite matrix and  $\gamma()$  and  $\rho()$  are positive semidefinite functions.



# $MA(q)$ processes

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## *Definition*

$MA(q)$ , a moving average process of order  $q$  is defined as

$$X_t = W_t + \theta_1 W_{t-1} + \cdots + \theta_q W_{t-q}$$



# Moving average process (MA)

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## Definition

A time series  $\{X_t\}$  is said to be an moving average process (of order one) if  $X_t = W_t + \theta W_{t-1}$  where  $W_t \sim WN(0, \sigma_w^2)$

- MA(1) process is a linear process with  $\mu = 0$  and

$$\psi_j = \begin{cases} 1 & \text{if } j = 0 \\ \theta & \text{if } j = 1 \\ 0 & \text{otherwise} \end{cases}$$

- $E(X_t) = 0$

■

$$\gamma_X(h) = \begin{cases} \sigma_w^2(1 + \theta^2) & \text{if } h = 0 \\ \sigma_w^2\theta & \text{if } h = +1, -1 \\ 0 & \text{otherwise} \end{cases}$$

- MA(1) is a at least weakly stationary process



# $AR(p)$ processes

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## *Definition*

$AR(p)$ , an auto-regressive process of order  $p$  is defined as

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + W_t$$



# Auto regressive process (AR)

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## Definition

A time series  $\{X_t\}$  is said to be an auto regressive process (of order one) if  $X_t = \phi X_{t-1} + W_t$  where  $W_t \sim WN(0, \sigma_w^2)$

- AR(1) process is a linear process with  $\mu = 0$  and

$$\psi_j = \begin{cases} 1 & \text{if } j = 0 \\ \phi^j & \text{if } j \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

- $E(X_t) = 0$
- $\gamma_X(h) = \frac{\sigma_w^2 \phi^{|h|}}{1 - \phi^2}$  if  $|\phi| < 1, \phi \neq 0$
- If  $\phi = 0$  then AR(1) process is a WN process
- AR(1) is at least a weakly stationary process

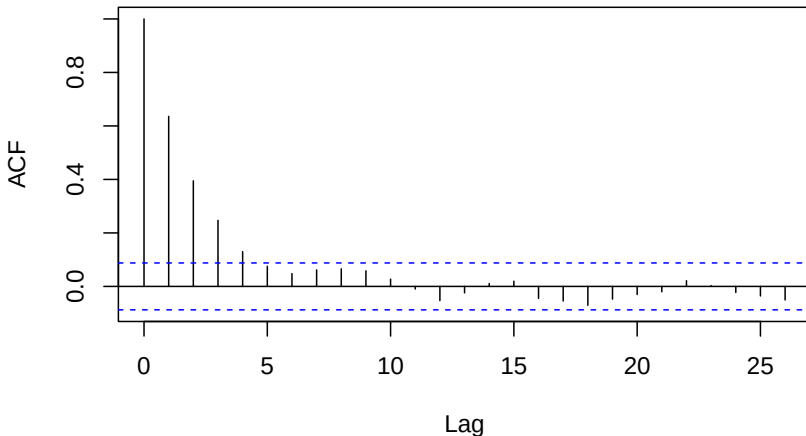
[exponential decay]



## AR(1) Example

```
set.seed(123); n<-500; p<-1; d<-0;q<-0;  
ar1.sim<-arima.sim(list(order=c(p,d,q), ar=0.7), n)  
acf(ar1.sim,type = "correlation",plot = T)
```

Series ar1.sim





# Converges in mean square

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## Definition

A sequence of random variables  $Y_1, Y_2, \dots$  converges in mean square to  $Z$  if for which

$$\lim_{n \rightarrow \infty} E(Y_n - Z)^2 = 0$$

- Consider AR(1) process  $X_t = \phi X_{t-1} + W_t$  where,  $W_t \sim WN(0, \sigma_w^2)$  and  $|\phi| < 1$ .
- AR(1) is an  $MA(\infty)$  process i.e.

$$\lim_{k \rightarrow \infty} E \left( X_t - \sum_{j=0}^k \phi^j W_{t-j} \right)^2 = 0$$

- Convergence in mean square  $\implies$  Convergence in probability  $\implies$  Convergence in distribution, BUT NOT THE OTHER WAY ROUND IN GENERAL.





## Use of sample ACF

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| Time series | ACF                          |
|-------------|------------------------------|
| White Noise | Zero                         |
| MA( $q$ )   | Zero for $ h  > q$           |
| AR( $p$ )   | Decays to zero exponentially |



## Definition

A linear process  $X_t$  is causal function of  $W_t$  if

$$X_t = \left( 1 + \sum_{i=1}^{\infty} \phi_i B^i \right) W_t$$

where  $\sum_{i=1}^{\infty} |\phi_i| < \infty$

- AR(1) i.e.  $X_t = \phi X_{t-1} + W_t$  is casual iff  $|\phi| < 1$  i.e.  $1 - \phi z$  has a solution out of the unit circle on complex plane  $\mathbb{C}$ .



## Definition

A linear process  $X_t$  is invertible function of  $W_t$  if there

$$W_t = \left( 1 + \sum_{i=1}^{\infty} \theta_i B^i \right) X_t$$

where  $\sum_{i=1}^{\infty} |\theta_i| < \infty$

- MA(1) i.e.  $X_t = W_t + \theta W_{t-1}$  is invertible iff  $|\theta| < 1$  i.e.  $1 + \theta z$  has a solution out of the unit circle on complex plane  $\mathbb{C}$ .



# *Auto regressive-Moving average process (ARMA (p,q))*

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## *Definition*

An **ARMA(p,q)** process  $X_t$  is a stationary process that satisfies

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = W_t + \theta_1 W_{t-1} + \cdots + \theta_q W_{t-q}$$

where  $W_i \sim WN(0, \sigma_w^2)$ .

- $AR(p) = ARMA(p, 0)$
- $MA(q) = ARMA(0, q)$



# ARMA( $p, q$ )

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- Back-shift operator  $B$ , such that  $BX_t = X_{t-1}$
- $\Phi_p(z) = 1 - \sum_{i=1}^p \phi_i z^i$  if  $\phi_p \neq 0$
- $\Theta_q(z) = 1 + \sum_{i=1}^q \theta_i z^i$  if  $\theta_q \neq 0$
- Then ARMA( $p, q$ ) model can be written as

$$\Phi_p(B)X_t = \Theta_q(B)W_t$$

if  $\Phi_p(z)$  and  $\Theta_q(z)$  have no common factor i.e. not a lower order ARMA model is possible.



# ARMA (1,1)

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$$X_t - \phi X_{t-1} = W_t + \theta W_{t-1} \quad \text{with } |\phi| < 1$$

- $E(X_t) = 0$

- 

$$\gamma(h) = \begin{cases} \sigma^2 \left[ 1 + \frac{(\theta + \phi)^2}{1 - \phi^2} \right] & \text{if } h = 0 \\ \sigma^2 \left[ \theta + \phi + \frac{\phi(\theta + \phi)^2}{1 - \phi^2} \right] & \text{if } h = 1 \\ \gamma(1)\phi^{h-1} & \text{if } h \geq 2 \end{cases}$$



# *ARMA(p,q): Stationarity, causality, and invertibility*

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## *Theorem*

*If  $\Phi_p$  and  $\Theta_q$  have no common factors, a (unique) stationary solution to  $\Phi_p(B)X_t = \Theta_q(B)W_t$  exists iff the roots of  $\Phi_p(z)$  avoid the unit circle i.e.*

$$|z| = 1 \implies \Phi_p(z) \neq 0.$$

- This ARMA(p,q) process is causal iff the roots of  $\Phi_p(z)$  are outside the unit circle i.e.  $|z| \leq 1 \implies \Phi_p(z) \neq 0$ .
- It is invertible iff the roots of  $\Theta_q(z)$  are outside the unit circle i.e.  $|z| \leq 1 \implies \Theta_q(z) \neq 0$ .



# Partial Auto correlation function (PACF)

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## Definition

Partial Auto correlation function  $\alpha(h)$  between  $X_h$  and  $X_0$  for given  $X_1, \dots, X_{h-1}$  the correlation between the linear prediction errors  $X_h - \text{Ln}(X_h|X_1, \dots, X_{h-1})$  and  $X_0 - \text{Ln}(X_0|X_1, \dots, X_{h-1})$

$$\alpha(h) = \frac{\gamma(h) - \tilde{\gamma}_{h-1}^T(1)\Gamma_{h-1}^{-1}\gamma_{h-1}(1)}{\gamma(0) - \gamma_{h-1}^T(1)\Gamma_{h-1}^{-1}\gamma_{h-1}(1)}$$

where

$$\tilde{\gamma}_{h-1}^T(1) = (\gamma(h-1)^T, \dots, \gamma(1)^T)$$

and

$$\gamma_{h-1}^T(1) = (\gamma(1), \dots, \gamma(h-1))^T$$

and

$$\Gamma_{h-1} = ((\gamma(|i-j|)))_{ij}$$





# Classification

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| Model:    | ACF:             | PACF:            |
|-----------|------------------|------------------|
| AR(p)     | decays           | zero for $h > p$ |
| MA(q)     | zero for $h > q$ | decays           |
| ARMA(p,q) | decays           | decays           |



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## Sample MEAN, ACVF & ACF

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For observations  $x_1, \dots, x_n$  of a time series,

$$\text{Sample mean } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The sample autocovariance function is

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-|h|} (x_t - \bar{x})(x_{t+|h|} - \bar{x}) \quad \text{if } -n < h < n$$

The sample auto-correlation function is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$



## About sample mean $\bar{x}$

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- $E(\bar{x}) = \mu$

[unbiased estimator]

- $Var(\bar{x}) = \frac{1}{n} \sum_{h=-n}^n \left(1 - \frac{|h|}{n}\right) \gamma(h) \rightarrow 0 \text{ as } n \rightarrow \infty$

- $\widehat{Var}(\bar{x}) = \frac{1}{n} \sum_{h=-n}^n \left(1 - \frac{|h|}{n}\right) \widehat{\gamma}(h)$

### Definition

Long run variance

$$\lim_{n \rightarrow \infty} nVar(\bar{x}) = \lim_{n \rightarrow \infty} \sum_{h=-\infty}^{\infty} \left(1 - \frac{|h|}{n}\right) \gamma(h) = \sigma_w^2 \left( \sum_{j=-\infty}^{\infty} \psi_j \right)^2$$

- $nVar(\bar{x}) \rightarrow \sum_{-\infty}^{\infty} \gamma(h)$  if  $\sum_{-\infty}^{\infty} |\gamma(h)| < \infty$  as  $n \uparrow \infty$ .



## About sample mean $\bar{x}$

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If  $\{X_t\}$  is a stationary Gaussian sequence then,

$$\sqrt{n}(\bar{x} - \mu) \sim N \left( 0, \sum_{|h| < n} \left(1 - \frac{|h|}{n}\right) \gamma(h) \right)$$

■ 95% CI of  $\mu$  is

$$\left( \bar{x} - 1.96 \frac{\sigma_n}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma_n}{\sqrt{n}} \right)$$

where  $\sigma_n^2 = \sum_{|h| < n} \gamma(h) \approx \sum_{|h| < \sqrt{n}} \hat{\gamma}(h)$



# Bartlett's formula about $\hat{\rho}(h)$

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If  $X_t = \mu + \sum_{j=-\infty}^{j=+\infty} \psi_j W_{t-j}$  with  $E(X_t^4) < \infty$  then the approximate joint density of  $(\hat{\rho}(i), \hat{\rho}(j))$  is

$$\begin{pmatrix} \hat{\rho}(i) \\ \hat{\rho}(j) \end{pmatrix} \sim N \left[ \begin{pmatrix} \rho(i) \\ \rho(j) \end{pmatrix}, \frac{1}{n} \begin{pmatrix} v_{ii} & v_{ij} \\ v_{ji} & v_{jj} \end{pmatrix} \right]$$

where,

$$v_{ij} = \sum_{h=1}^{\infty} (\rho(h+i) + \rho(h-i) - 2\rho(i)\rho(h)) \times (\rho(h+j) + \rho(h-j) - 2\rho(j)\rho(h))$$



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# Linear Forecasting : Durbin-Levinson algorithm

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- Given  $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$  the best linear predictor

$$\hat{X}_{m+n}^n = \sum_{i=1}^n \alpha_i X_i$$

satisfying the following conditions

$$(1) \quad E(\hat{X}_{m+n}^n - X_{m+n}) = 0 \quad [\text{Unbiased prediction}]$$

$$(2) \quad E[(\hat{X}_{m+n}^n - X_{m+n})X_i] = 0 \quad [\text{Error is orthogonal to predictors}]$$

- Durbin-Levinson estimate : Coefficient for 1 step prediction

$$\hat{\alpha} = \Gamma_n^{-1} \gamma_n(1)$$

- Prediction error:  $E(X_{n+1} - \hat{\alpha}^T \mathbf{X})^2 = \gamma(0) - \gamma_n^T(1) \Gamma_n^{-1} \gamma_n(1)$





# Estimation of Trend in the Absence of Seasonality

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- Smoothing with a finite moving average filter: Let  $q$  be a nonnegative integer and consider the two-sided moving average

$$m_t = \frac{1}{2q+1} \sum_{i=t-q}^{t+q} X_i \approx \frac{1}{2q+1} \sum_{i=t-q}^{t+q} T_i \approx T_t$$

if  $T_t$  is linear in  $[t-q, t+q]$

- Linea filter :

$$m_t = \sum_{i=-q}^q a_i X_{t-i}$$



# Estimation of Trend in the Absence of Seasonality

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- Exponential smoothing: For any fixed  $\alpha \in [0, 1]$ , the one-sided moving  $m_t = X_1$  and

$$m_t = \alpha X_t + (1 - \alpha)m_{t-1} \quad \text{for } t = 2, \dots, n$$

- Polynomial fitting: For example fit  $m_t = a_0 + a_1 t + a_2 t^2$  which minimize the square distance  $\sum_{t=1}^n (x_t - m_t)^2$ .
- Trend Elimination by Differencing: Estimate  $k$  such that  $\nabla^k X_t \approx \text{constant}$ . Then fit  $k$ -degree polynomial



# Estimation of Trend and Seasonality

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- Additive model :  $X_t = T_t + S_t + W_t$

with  $E(W_t) = 0$ ,  $S_{t+d} = S_t$  and  $\sum_{t=1}^d S_t = 0$  then

$$\nabla_d X_t = T_t - T_{t-d} + W_t - W_{t-d}$$

- Now the trend  $T_t - T_{t-d}$  can be removed by using any of the above methods.



# Linear Forecasting : Durbin-Levinson algorithm

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- Given  $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$  the best linear predictor

$$\hat{X}_{m+n}^n = \sum_{i=1}^n \alpha_i X_i$$

satisfying the following conditions

$$(1) \quad E(\hat{X}_{m+n}^n - X_{m+n}) = 0 \quad [\text{Unbiased prediction}]$$

$$(2) \quad E[(\hat{X}_{m+n}^n - X_{m+n})X_i] = 0 \quad [\text{Error is orthogonal to predictors}]$$

- Durbin-Levinson estimate : Coefficient for 1 step prediction

$$\hat{\alpha} = \Gamma_n^{-1} \gamma_n(1)$$

- Prediction error:  $E(X_{n+1} - \hat{\alpha}^T \mathbf{X})^2 = \gamma(0) - \gamma_n^T(1) \Gamma_n^{-1} \gamma_n(1)$



## Estimation of Model parameters (Yule-Walker)

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- Remove the trend
- Remove the seasonal effect
- Now for weakly stationary process use

(1) Yule-Walker estimator : This eventually a method of moment estimation process. So equate the theoretical moments with the corresponding sample moments of ARMA(p,q) and solve them.

■

$$\hat{\Gamma}_n \hat{\alpha} = \hat{\gamma}_n(1)$$

■

$$\hat{\sigma}^2 = \hat{\gamma}(0) - \hat{\alpha}^T \hat{\gamma}_n(1)$$



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# Integrated ARMA Models: $ARIMA(p, d, q)$

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## Definition

For  $p, d, q \geq 0$ , we say that a time series  $X_t$  is an  $ARIMA(p, d, q)$  process if  $Y_t = (1 - B)^d X_t$  is  $ARMA(p, q)$ . We can write

$$\Phi_p(B) \nabla^d X_t = \Theta_q(B) W_t.$$

- Example: [  $ARIMA(0,1,0)$  ] Random walk with drift

$$X_t = \mu t + \sum_{i=0}^t W_i \quad \text{where } W_i \sim N(0, \sigma^2) \text{ i.i.d.}$$

This implies

$$\nabla X_t - \mu \sim N(0, \sigma^2) \quad \text{which is a White noise}$$



The Akaike information criterion (AIC) is an estimator of the relative quality of statistical models for a given set of data. Given a collection of models for the data, AIC estimates the quality of each model, relative to each of the other models. Thus, AIC provides a means for model selection.

Suppose that we have a statistical model of some data. Let  $k$  be the number of estimated parameters in the model. Let  $\hat{L}$  be the maximum value of the likelihood function for the model. Then the AIC value of the model is the following.

$$AIC = 2k - 2\ln(\hat{L})$$





In statistics, the Bayesian information criterion (BIC) is a criterion for model selection among a finite set of models; the model with the lowest BIC is preferred.

The BIC is formally defined as

$$\text{BIC} = \ln(n)k - 2 \ln(\hat{L}).$$